

CHAPTER 8 : USEFUL AND HARMFUL MICROBES

Topic:-

Collect information about generic medicines and discuss them in the class.

Introduction:-

Healthcare is a fundamental human right. However, for millions of low-income and middle-class families in India and worldwide, the exorbitant cost of prescription pharmaceuticals poses a severe economic burden. In many instances, life-saving treatments for chronic ailments like diabetes, hypertension, and bacterial infections remain unaffordable solely due to inflated drug branding.

To tackle this healthcare challenge, modern medicine relies increasingly on **Generic Medicines**. A generic medicine is a pharmaceutical drug that contains the exact same active chemical ingredient (Active Pharmaceutical Ingredient - API), in the identical dosage, strength, safety profile, quality, route of administration, and therapeutic effectiveness as the original patent-holding branded medicine.

For this project, we conducted extensive investigative research into generic pharmaceuticals, visited local pharmacies and a government Pradhan Mantri Bhartiya Janaushadhi Kendra, compiled comprehensive price-comparison data sheets, and led an engaging scientific discussion in our science classroom.

❖ Core Characteristics of Generic Medicines:-

- Identical in chemical formula, purity, bioequivalence, and curative potency to branded drugs.
- Sold under their official chemical international non-proprietary name (INN) rather than a brand name.
- Priced 50% to 90% cheaper because manufacturers incur zero patent development or advertising expenses.

Aim & Investigation Setup:-

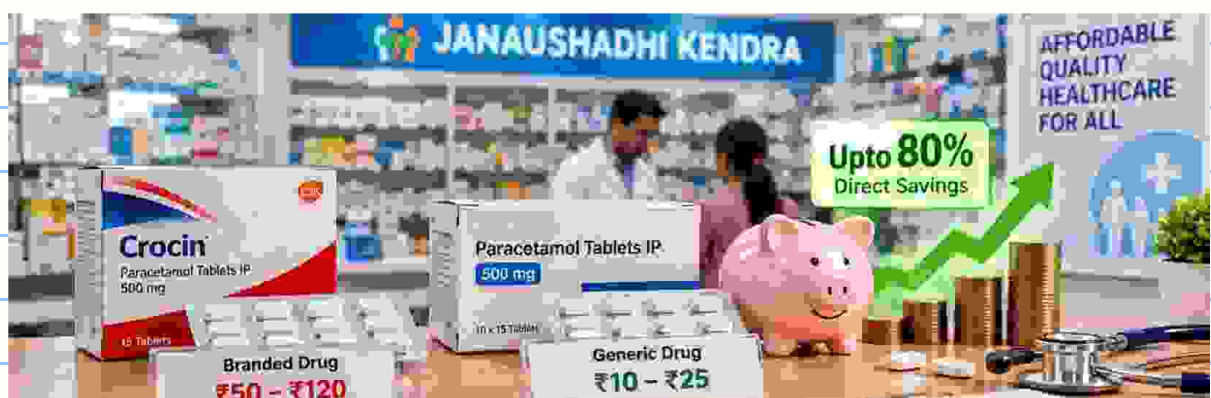
Aim: To research the pharmaceutical, regulatory, and economic realities of generic medicines, gather comparative price data from pharmacies, and facilitate a classroom discussion on affordable healthcare.

❖ Objectives of the Project:-

- To understand the scientific definition of Active Pharmaceutical Ingredients (APIs) and bioequivalence.
- To investigate why branded drugs are expensive and how generic medicines drastically cut healthcare bills.
- To visit a Janaushadhi Kendra and compare retail prices of five widely prescribed essential drugs.
- To address common public misconceptions about generic medicines during a class discussion.

❖ Resources & Data Sources Consulted:-

- Interview with registered pharmacist Shri Anil Deshmukh at a local Janaushadhi Kendra.
- Official price list published by the Department of Pharmaceuticals, Ministry of Chemicals & Fertilizers.
- Class 9 Science textbook (Chapter 8: Useful and Harmful Microbes - Antibiotics section).
- Sample empty packaging strips of branded Paracetamol, Metformin, and their generic equivalents.



Survey Methodology & Pharmacy Visit:-

Our field study and data collection were carried out through five systematic phases:

❖ *Field Survey & Research Methodology:-*

- **Phase 1 (Literature & Regulatory Review):** Studied the Indian Drugs and Cosmetics Act (1940) and guidelines set by the Central Drugs Standard Control Organisation (CDSCO) regarding bioequivalence testing.
- **Phase 2 (Pharmacy Field Visits):** Visited two commercial private retail pharmacies and one government-sponsored Pradhan Mantri Bhartiya Janaushadhi Kendra in our municipal ward.
- **Phase 3 (Price & Ingredient Auditing):** Audited five essential everyday medications prescribed for fever, diabetes, bacterial infection, acidity, and allergic rhinitis. Recorded maximum retail prices (MRP) and Active Pharmaceutical Ingredient (API) concentrations.
- **Phase 4 (Interview with Pharmacist):** Interviewed the pharmacist regarding supply chain logistics, quality testing certifications (WHO-GMP standards), and public purchasing patterns.
- **Phase 5 (Class Presentation & Debate):** Prepared a comparative chart board and conducted a 20-minute structured presentation followed by open Q&A in our classroom.

Observations & Price Comparison Table:-

Our direct price audit revealed shocking disparities between heavily advertised branded drugs and their certified generic equivalents:

Active Ingredient (API)	Common Branded Name	Branded Price (Strip)	Generic Price (Strip)	Consumer Savings (%)
Paracetamol (650 mg)	Dolo-650 / Calpol	₹34.00	₹6.50	80.8% Savings
Amoxicillin (500 mg)	Novamox / Mox	₹115.00	₹28.00	75.6% Savings
Metformin (500 mg)	Glycomet	₹48.00	₹8.50	82.3% Savings
Omeprazole (20 mg)	Omez	₹65.00	₹12.00	81.5% Savings
Cetirizine (10 mg)	Cetzine	₹24.00	₹4.50	81.2% Savings
Atorvastatin (10 mg)	Atorva / Lipitor	₹145.00	₹22.00	84.8% Savings

In every single medicine examined, generic equivalents delivered savings exceeding 75% while providing identical milligrams of active drug.

Scientific Analysis & Why Generics Cost Less:-

During our classroom presentation, we explained the scientific and economic reasons behind the vast price differences:

❖ Why Branded Drugs are Expensive vs Why Generics are

Affordable
High Cost of Research & Development (R&D) Costs: Developing a novel drug molecule takes over 10 to 15 years of chemical synthesis and clinical trials, costing hundreds of millions of dollars. Once the innovator's 20-year patent expires, generic manufacturers can legally synthesize the proven molecule without duplicating R&D costs.

- **Zero Marketing & Promotional Expenses:** Branded drug firms spend heavily on medical representative (MR) visits, glossy brochures, brand advertising, and physician gifts. Generic producers spend almost nothing on brand promotions, passing savings directly to buyers.

- **Stringent Bioequivalence & WHO-GMP Standards:** Regulatory bodies (CDSCO in India, US FDA globally) mandate that generic drugs must prove *bioequivalence*—meaning the rate and extent of drug absorption in human blood must match the branded drug within strict limits ($\pm 5\%$).

Busting Common Myths & Classroom Discussion:-

During the interactive session, our classmates raised common doubts which we systematically clarified based on scientific facts:

❖ Debunking Prevalent Healthcare Myths:-

- **Myth 1: "Cheap medicines mean poor or inferior quality."**

Scientific Reality: Low cost stems solely from bypassed marketing and patent expiry, not substandard chemistry. Generic drugs produced in WHO-GMP certified facilities undergo rigorous chemical purity and dissolution testing.

- **Myth 2: "Generic medicines take longer to work."**

Scientific Reality: Because bioequivalence is legally guaranteed, the active chemical molecules dissolve, enter the bloodstream, and cure infections at the identical therapeutic rate.

- **Myth 3: "Doctors always write the best medicine."**

Scientific Reality: Many doctors write trade brand names out of habit or aggressive marketing. The National Medical Commission (NMC) now legally urges doctors to prescribe medicines using generic international names.

❖ Role of Government Initiatives: Pradhan Mantri Janaushadhi Kendras offer over 1,800 generic medicines across India.

What We Learned, Conclusion & References:-

❖ What We Learned From This Project:-

- Understood the pharmaceutical concept of bioequivalence and Active Pharmaceutical Ingredients.
- Learned how economic mechanisms govern medicine pricing and the transformative power of generic drugs.
- Conducted meaningful field interviews, compiled analytical price audits, and led peer discussions.

❖ Project Conclusion:-

Generic medicines represent an indispensable scientific and humanitarian tool to make healthcare affordable and accessible for all citizens. Our study conclusively proved that generic medicines provide identical therapeutic efficacy and safety as branded drugs at a fraction of the cost. Spreading awareness about Janaushadhi Kendras and asking pharmacists for generic equivalents can save families thousands of rupees every year.

❖ Bibliography & References:-

- Maharashtra State Bureau of Textbook Production (Balbharati) – Standard 9 Science & Technology (Chapter 8: Useful and Harmful Microbes).
- Pharmaceuticals & Medical Devices Bureau of India (PMBI) – Official Janaushadhi Product & Price Guide.
- World Health Organization (WHO) – Guidelines on Bioequivalence and Essential Generic Medicines.